

EDUCATION	
University of Waterloo <i>Bachelor of Computer Engineering, Co-op</i>	Expected April 2026 Waterloo, ON
SKILLS	
Languages: Go, Java, C, C++, Rust, Python, Swift, Protobuf, Lua, Bash, SQL, SystemVerilog, TLA+ Tools: OpenAI Codex, Claude Code, MCP, Git(Worktree), Jira, Docker, Postman Systems: NoSQL(MongoDB), RDBMS, Redis, etcd, Kubernetes, Flink, Kafka, Spark, TensorFlow, PyTorch	
EXPERIENCE	
Apple Inc. <i>Software Engineering Intern – Apple Intelligence Infrastructure, Vector Database, Swift, SQLite</i>	Sept. 2025 – Dec. 2025 Waterloo, ON
- Contributed to <u>MicroNN</u>, Apple’s on-device, ANN vector database used by numerous iOS/macOS system apps. - Lowered p95 search query latency by 90%, by implementing a budget-based IVF ANN search algorithm in Swift. - Performed deep CPU flamegraph profiling, syscall/I/O tracing, and end-to-end tail latency decomposition to optimize SQLite-backed meta-data access and eliminate cold-start bottlenecks. - Delivered a backward-compatible API addition adopted by multiple internal clients (Photos, Mail) across iOS.	
Kuzu Inc. <i>Software Engineering Intern – Database Internals (KuzuDB)</i>	Aug. 2025 – Sept. 2025 Waterloo, ON
Introduced a randomized seed-driven test framework to stress-test the transactional system’s invariants (isolation, atomicity, commit ordering), improving reproducibility of rare concurrency bugs and saving tens of engineering hours.	
Cockroach Labs <i>Software Engineering Intern – Core Distributed Systems Internals, Golang</i>	Jan. 2025 – Apr. 2025 New York City, NY
- Worked on optimizing CockroachDB’s core Raft distributed consensus algorithm implementation. - Engineered a data structure for Raft’s replicated log in Go, reducing store-wide cache miss rate by 20%. - Optimized Raft consensus by re-writing its log conflict resolution algorithm and pipelining it with leader election. - Reducing Raft’s post-election replication latency by up to 50% (1 RTT). - Eliminated up to 3 seconds of client-visible database unavailability during extreme fail-overs. - Developed deep understanding in hyperscale distributed systems, topics including: maintainability, monitoring, fault tolerance, load balancing, clock synchronization, distributed transactions, zero-downtime rollouts, multi-tenancy, and distributed algorithms (Raft/Paxos, ZooKeeper, Gossip, 2PC, Percolator). - My contributions: github.com/cockroachdb/cockroach/commits?author=hakuuww	
Auvik Networks <i>Software Engineering Intern – Backend, Event-Driven Stream Processing, Golang, Kafka, Flink, K8S</i>	May 2024 – Aug. 2024 Waterloo, ON
- Maintained a real-time stream processing pipeline using Kafka and Flink coupled with Java & Go microservices, processing 20k+ events per second with sub-second end-to-end latency. - Designed a solution for Kafka large file passing using AWS S3, balancing fault tolerance, latency and throughput. - Leveraged Helm and YAML to templatize Kubernetes deployments, streamlining multi-environment rollouts and boosting deployment efficiency by 20%.	
Government of Ontario <i>Software Engineering Intern - Backend, Database Admin., Oracle Relational Database</i>	Jan. 2023 – Apr. 2023 Toronto, ON
Developed batched ETL operations using Bash, C#, .NET and PL/SQL.	
Teranet <i>Software Engineering Intern - Backend</i>	May. 2022 – Aug. 2022 Toronto, ON
Aided the launch of a government-approved software solution by performing automated Selenium Web testing in C#, identifying and resolving 15 critical bugs, ensuring successful production deployment.	
PROJECTS	
Object Storage on Discord <i>Go, Gin, REST API Design, MongoDB NoSQL (metadata store), Docker</i> File storage using chunked distribution across Discord channels with a NoSQL catalog for indexing and retrieval.	
MapReduce <i>Go, MIT 6.5840 Distributed Systems</i> Developed a single-machine MapReduce from scratch; distributed nodes are simulated by individual goroutines.	
TECHNICAL DEEP DIVES (PAPERS & SYSTEMS)	
MapReduce, GFS, BigTable, Spanner, Dapper, Borg, Monarch(Google), Raft, Paxos, etcd, Temporal, FoundationDB, KuzuDB CockroachDB(2020/2022/2025), MicroNN ANN vector database (Apple), DynamoDB, Aurora(Amazon), Spark RDD, Kafka	